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APPARATUS AND METHOD FOR CALCULATING INTERNAL RESISTANCE OF BATTERY, AND COMPUTER PROGRAM

Abstract

The present disclosure relates to an apparatus and method for calculating an internal resistance of a battery, and a non-transitory computer-readable medium. Some embodiments provide a topology capable of measuring an internal resistance of a battery cell even when the battery is mounted on a vehicle without the separation or removal of the battery from the vehicle. Some embodiments of the present disclosure provide a configuration for calculating an internal resistance of a battery based on a current parameter and a voltage parameter of the battery when a data collection condition is satisfied, the data collection condition being predefined based on the discharge current characteristics of the battery while a vehicle is in a driving state.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit under 35 U.S.C. § 119 (a) of Korean Patent Application No. 10-2024-0031334, filed on Mar. 5, 2024, the disclosure of which is incorporated herein by reference in its entirety.

FIELD

[0002] The present disclosure relates to an apparatus and method for calculating an internal resistance (direct current internal resistance (DCIR)) of a battery, and a non-transitory computer-readable medium.

BACKGROUND

[0003] A battery pack is manufactured through a method in which single cells are welded and accommodated in a housing to assemble modules, and then the modules are connected through busbars and wires. Since a process of storing (charging) and consuming (discharging) electrical energy is repeated, it is very important to secure data on resistance values which are directly related to the lifespan. In particular, in order to predict and manage the lifespan due to repeated charging or discharging of a battery pack mounted on an electric vehicle, it is necessary to accurately and easily calculate the internal resistance (direct current internal resistance (DCIR)) of a battery cell.

[0004] Generally, the internal resistance of a battery cell is calculated based on an amount of voltage drop measured in a process of discharging a step current through a charger/discharger. A method of measuring an internal resistance of a battery cell mounted on an electric vehicle is not standardized, but generally, a method of discharging a battery cell for a certain time (10 seconds) through a charger/discharger and calculating an internal resistance through a voltage drop thereof is applied.

[0005] However, such a method is based on the internal resistance of a battery cell being measured in a state in which the battery cell is separated from a vehicle, and since the battery cell cannot be separated in a driving state of the vehicle, there are temporal and spatial limitations in measuring the internal resistance of the battery cell.

[0006] The above information disclosed in this Background section is for enhancement of understanding of the background of the present disclosure, and therefore, it may contain information that does not constitute related (or prior) art.

SUMMARY

[0007] The present disclosure is directed to providing an apparatus and method for calculating an internal resistance of a battery, and a non-transitory computer readable medium, in which, by measuring an internal resistance of a battery cell mounted on a vehicle while the vehicle is in a driving state (e.g., an actual driving state or a simulated driving state), without the separation or removal of the battery from the vehicle, the internal resistance of the battery cell can be measured even when the battery is mounted on the vehicle.

[0008] However, objects that the present disclosure intends to achieve are not limited to the above-described objects and other objects that are not described may be clearly understood by those skilled in the art from the following description.

[0009] According to an aspect of the present disclosure, an apparatus for calculating an internal resistance of a battery is provided, the apparatus including a processor and a memory configured to store instructions for execution by the processor, wherein, when executing the instructions, the processor is configured to calculate an internal resistance of a battery mounted on a vehicle by, when the processor determines that a data collection condition is satisfied, collecting a current

parameter and a voltage parameter of the battery and calculating the internal resistance of the battery based on the collected current parameter and voltage parameter, the data collection condition being predefined based on discharge current characteristics of the battery while the vehicle is in a driving state.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The following drawings accompanying this specification illustrate some embodiments of the present disclosure, and further describe some aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed as being limited to the drawings. In the drawings:

[0011] FIG. 1 illustrates an example prismatic battery according to one embodiment of the present disclosure;

[0012] FIG. 2 is a cross-sectional view illustrating the prismatic battery of FIG. 1 according to one embodiment of the present disclosure;

[0013] FIG. 3 illustrates an example battery module according to one embodiment of the present disclosure;

[0014] FIGS. 4A and 4B illustrate an example battery pack according to one embodiment of the present disclosure;

[0015] FIGS. 5A and 5B illustrate an example vehicle body and vehicle body parts according to one embodiment of the present disclosure;

[0016] FIG. 6 is a block diagram of an example apparatus for calculating an internal resistance of a battery according to one embodiment of the present disclosure;

[0017] FIG. 7 shows an example of a data collection condition in an apparatus for calculating an internal resistance of a battery according to one embodiment of the present disclosure.

[0018] FIGS. 8 to 10 show an example process for calculating an internal resistance in the apparatus for calculating an internal resistance of a battery according to one embodiment of the present disclosure; and

[0019] FIGS. 11 to 13 are a flowchart of an example method of calculating an internal resistance of a battery according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

[0020] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in this specification and claims should not be construed as being limited to the usual or dictionary meaning and should be interpreted as meaning and concept consistent with the technical idea of the present disclosure based on the principle that the inventor can be his/her own lexicographer to appropriately define the concept of the term to explain his/her disclosure in the best way.

[0021] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical ideas, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0022] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first

element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements. [0023] In the figures, dimensions of the various elements, layers, etc., may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B and C,” “at least one of A, B or C,” “at least one selected from a group of A, B and C,” or “at least one selected from among A, B and C” are used to designate a list of elements A, B and C, the phrase may refer to any and all suitable combinations or a subset of A, B and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0024] It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0025] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

[0026] The terminology used herein is for the purpose of describing embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0027] Also, any numerical range disclosed and/or recited herein is intended to include all subranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant

reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein.

[0028] References to two compared elements, features, etc., as being “the same” may mean that they are “substantially the same”. Thus, the phrase “substantially the same” may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.

[0029] Throughout the specification, unless otherwise stated, each element may be singular or plural.

[0030] When an arbitrary element is referred to as being disposed (or located or positioned) on the “above (or below)” or “on (or under)” a component, it may mean that the arbitrary element is placed in contact with the upper (or lower) surface of the component and may also mean that another component may be interposed between the component and any arbitrary element disposed (or located or positioned) on (or under) the component.

[0031] In addition, it will be understood that when an element is referred to as being “coupled,” “linked” or “connected” to another element, the elements may be directly “coupled,” “linked” or “connected” to each other, or an intervening element may be present therebetween, through which the element may be “coupled,” “linked” or “connected” to another element. In addition, when a part is referred to as being “electrically coupled” to another part, the part can be directly connected to another part or an intervening part may be present therebetween such that the part and another part are indirectly connected to each other.

[0032] Before a detailed description of the present embodiments, first, the structure of an example battery that is a target of which an internal resistance is to be measured in the present embodiment will be briefly described.

[Prismatic Battery]

[0033] FIG. 1 is a perspective view illustrating a secondary battery to which embodiments of the present disclosure may be applied, and FIG. 2 is a cross-sectional view of the secondary battery along line II-II in FIG. 1.

[0034] Referring to FIGS. 1 and 2, the secondary battery **100** according to one or more embodiments of the present disclosure may include at least one electrode assembly **10** wound with a separator **13** as an insulator between the positive electrode **11** and the negative electrode **12**, a case **20** in which the electrode assembly **10** is received (or accommodated), and a cap assembly **30** coupled to an opening of the case **20**.

[0035] The secondary battery **100** according to one or more embodiments illustrated in FIGS. 1 and 2 will now be described as an example of a prismatic lithium-ion secondary battery. However, the present disclosure is not limited thereto, and suitable aspects, features and principles described herein may be applied to various other types of batteries, such as lithium polymer batteries and/or cylindrical batteries.

[0036] Each of the positive electrode **11** and the negative electrode **12** may include a current collector made of a thin metal foil having a coated portion on which an active material is coated and an uncoated portion **11a**, **12a** on which an active material is not coated.

[0037] The positive electrode **11** and the negative electrode **12** are wound after interposing the separator **13**, which is an insulator, therebetween. However, the present disclosure is not limited thereto, and the electrode assembly **10** may have a structure in which a positive electrode **11** and a negative electrode **12**, each made of a plurality of sheets, are alternately stacked with a separator interposed therebetween.

[0038] The case **20** may form the overall outer appearance of the secondary battery **100** and may be made of a conductive metal, such as aluminum, aluminum alloy, or nickel-plated steel. In addition, the case **20** may provide a space in which the electrode assembly **10** is accommodated.

[0039] The cap assembly **30** may include a cap plate **31** covering an opening in the case **20**, and the

case **20** and the cap plate **31** may be made of a conductive material. The positive and negative electrode terminals **21** and **22**, electrically connected to the positive electrode **11** and the negative electrode **12**, respectively, may be installed to penetrate (or extend through) the cap plate **31** and protrude outwardly therethrough.

[0040] In addition, outer peripheral surfaces (e.g., circumferential surfaces) of upper pillars of the positive and negative electrode terminals **21** and **22**, protruding outwardly from the cap plate **31**, may be threaded and may be fixed to the cap plate **31** by utilizing nuts.

[0041] However, the present disclosure is not limited thereto, and the positive and negative electrode terminals **21** and **22** may have a rivet structure and may be riveted or welded to the cap plate **31**.

[0042] In addition, the cap plate **31** may be made of a thin plate and may be coupled to the opening in the case **20**, and an electrolyte injection port **32**, into which a sealing stopper **33** may be installed, may be located (e.g., formed) in the cap plate **31**, and a vent portion **34** having a notch **34a** may be installed.

[0043] The positive and negative electrode terminals **21** and **22** may be electrically connected to current collectors, including first and second current collectors **40** and **50** (hereinafter referred to as positive and negative current collectors), by being bonded or coupled (e.g., by welding) to the positive uncoated portion **11a** and the negative electrode uncoated portion **12a**, respectively.

[0044] For example, the positive and negative electrode terminals **21** and **22** may be coupled by welding to the positive and negative electrode current collectors **40** and **50**, respectively. However, the present disclosure is not limited thereto, and the positive and negative electrode terminals **21** and **22** and the positive and negative electrode current collectors **40** and **50** may be integrally formed in one or more embodiments.

[0045] In addition, an insulation member may be installed between the electrode assembly **10** and the cap plate **31**. The insulation member may include first and second lower insulation members **60** and **70**, and each of the first and second lower insulation members **60** and **70** may also have a portion located between the electrode assembly **10** and the case **20**.

[0046] In addition, according to one or more embodiments of the present disclosure, one end of a separation member may face one side of the electrode assembly **10** and may be installed between the insulation member and the positive or negative electrode terminals **21** and **22**.

[0047] In one or more embodiments, the separation member may include first and second separation members **80** and **90**.

[0048] In such an embodiment, first ends of the first and second separation members **80** and **90** installed to face one side of the electrode assembly **10** may be respectively installed between the first and second lower insulation members **60** and **70** and the positive and negative electrode terminals **21** and **22**.

[0049] Accordingly, the positive and negative electrode terminals **21** and **22**, which may be coupled by welding to the positive and negative electrode current collectors **40** and **50**, may be coupled to first ends of the first and second lower insulation members **60** and **70** and the first and second separation members **80** and **90**.

[Battery Module]

[0050] FIG. **3** is a perspective view illustrating an example battery module according to one embodiment of the present disclosure.

[0051] Referring to FIG. **3**, the battery module **300** according to one or more embodiments of the present disclosure includes terminal parts **311** and **312**, a plurality of battery cells **310** arranged in one direction, a connection tab **320** connecting a battery cell **310a** to an adjacent battery cell **310b**, and a protection circuit module **330** having one end connected to the connection tab **320**. The protection circuit module **330** may include a battery management system (BMS). Further, the connection tab **320** may include a body portion in contact with the terminal parts **311** and **312** between the adjacent battery cells **310a** and **310b** and an extension portion extending from the body

portion and connected to the protection circuit module **330**. The connection tab **320** may be, for example, a bus bar.

[0052] Each battery cell **310** may include a battery case, an electrode assembly received (or accommodated) in the battery case, and an electrolyte. The electrode assembly and the electrolyte react electrochemically to store and release (e.g., generate) energy. Terminal parts **311** and **312** electrically connected to the connection tab **320** and a vent **313** as a discharge passage for gas generated inside the battery case may be provided on one side of (e.g., an upper side of) the battery cell **310**. The terminal parts **311** and **312** of the battery cell **310** may be a positive electrode terminal **311** and a negative electrode terminal **312** having different polarities from each other, and the terminal parts **311** and **312** of the adjacent battery cells **310a** and **310b** may be electrically connected to each other in series or parallel by the connection tab **320**, to be described in more detail below. Although a serial connection has been described as an example, the connection structure is not limited thereto, and various connection structures may be employed as desired or necessary. In addition, the number and arrangement of battery cells is not limited to the structure shown in FIG. 3 and may be changed as desired or necessary.

[0053] The plurality of battery cells **310** may be arranged in (e.g., may be stacked in) one direction so that the wide surfaces of the battery cells **310** face each other, and the plurality of battery cells **310** may be fixed by housing components **361**, **362**, **363**, and **364**. The housing components **361**, **362**, **363**, and **364** may include a pair of end plates **361** and **362** facing the wide surfaces of the battery cells **310** and a side plate **363** and a bottom plate **364** connecting the pair of end plates **361** and **362** to each other. The side plate **363** may support side surfaces of the battery cells **310**, and the bottom plate **364** may support bottom surfaces of the battery cells **310**. In addition, the pair of end plates **361** and **362**, the side plate **363** and the bottom plate **364** may be connected by bolts **365** and/or any other suitable fastening members and methods known to those of ordinary skill in the art.

[0054] The protection circuit module **330** may have electronic components and protection circuits mounted thereon and may be electrically connected to connection tabs **320**, to be described in more detail later. The protection circuit module **330** includes a first protection circuit module **330a** and a second protection circuit module **330b** extending along the direction in which the plurality of battery cells **310** are arranged in different locations. The first protection circuit module **330a** and the second protection circuit module **330b** may be spaced from each other at a suitable interval (e.g., a predetermined interval) and arranged parallel to each other to be electrically connected to adjacent connection tabs **320**, respectively. For example, the first protection circuit module **330a** extends on one side of the upper portion of the plurality of battery cells **310** along the direction in which the plurality of battery cells **310** are arranged, and the second protection circuit module **330b** extends to the other upper side of the plurality of battery cells **310** along the direction in which the plurality of battery cells **310** are arranged. The second protection circuit module **330b** may be spaced from the first protection circuit module **330a** at a suitable interval (e.g., a predetermined interval) with the vents **313** interposed therebetween but may be disposed parallel to the first protection circuit module **330a**. As such, the two protection circuit modules are spaced from each other side-by-side and extend along the direction in which the plurality of battery cells **310** are arranged, thereby reducing or minimizing the area of the printed circuit board (PCB) constituting the protection circuit module. By separately configuring the protection circuit module into two protection circuit modules, unnecessary PCB area can be reduced or minimized. In addition, the first protection circuit module **330a** and the second protection circuit module **330b** may be connected to each other by a conductive connection member **350**. One side of the conductive connection member **350** is connected to the first protection circuit module **330a**, and the other side thereof is connected to the second protection circuit module **330b** so that the two protection circuit modules **330a** and **330b** can be electrically connected with each other.

[0055] The connection may be performed by any one of soldering, resistance welding, laser

welding, projection welding, and/or any other suitable connection methods known to those of ordinary skill in the art.

[0056] In addition, the connection member **350** may be, for example, an electric wire. In addition, the connection member **350** may be made of a material having elasticity or flexibility. By the connecting member **350**, it may be possible to check and manage whether the voltage, temperature, and/or current of the plurality of battery cells **310** are normal. For example, the information received by the first protection circuit module from connection tabs adjacent to the first protection circuit module **330a**, such as voltage, current, and/or temperature, and the information received from connection tabs adjacent to the second protection circuit module **330b**, such as voltage, current, and/or temperature, may be integrated and managed by the protection circuit module **330** through the connection member **350**.

[0057] In addition, when the battery cell **310** swells, shock may be absorbed by the elasticity or flexibility of the connection member **350**, thereby preventing the first and second protection circuit modules **330a** and **330b** from being damaged.

[0058] In addition, the shape and structure of the connection member **350** is not limited to the shape and structure shown in FIG. 3.

[0059] As described above, because the protection circuit module **330** is provided as the first and second protection circuit modules **330a** and **330b**, the area of the PCB constituting the protection circuit module can be reduced or minimized, and additional space inside the battery module can be obtained, which improves work efficiency by facilitating fastening to connect the connection tab **320** and the protection circuit module **330**, and repair work if (or when) an abnormality is detected in the battery module.

[Battery Pack]

[0060] FIGS. 4A and 4B are exemplary views illustrating an example battery pack **400** according to one exemplary embodiment of the present disclosure.

[0061] The battery pack **400** may include a plurality of battery modules **450** and a housing **410** for accommodating the plurality of battery modules **450**. For example, the housing **410** may include first and second housing components **411** and **412** coupled in opposite directions through the plurality of battery modules **450**. The plurality of battery modules **450** may be electrically connected to each other by using a bus bar **451**, and the plurality of battery modules **450** may be electrically connected to each other in a series/parallel or series-parallel mixed method, thereby obtaining desired (e.g., required) electrical output.

[Vehicle Body and Vehicle Body Parts]

[0062] FIGS. 5A and 5B are exemplary views illustrating an example vehicle body including a battery pack and vehicle body parts according to one exemplary embodiment of the present disclosure.

[0063] In FIG. 5A, a battery pack **591** may include a battery pack cover **513**, which is a part of a vehicle underbody **592**, and a pack frame **510** disposed under the vehicle underbody **592**. The pack frame **510** and the battery pack cover **513** may be integrally formed with a vehicle floor **582**. The vehicle underbody **592** separates the inside and outside of a vehicle, and the pack frame **510** may be disposed outside the vehicle.

[0064] FIG. 5B is a schematic side view of a vehicle according to one or more embodiments of the present disclosure. A vehicle **500** may be formed by combining additional parts, such as a hood **597** in front of the vehicle and fenders **598** respectively located in the front and rear of the vehicle to a vehicle body **599**. The vehicle **500** may further include a vehicle floor **582**, which is one of the vehicle body parts **590** including the battery pack **591** including the pack frame **510** and the battery pack cover **513**.

[0065] Hereinafter, based on such a battery structure, a specific method by which an internal resistance is calculated in the present embodiment will be described, focusing on an operation of a processor.

[0066] FIG. 6 is a block diagram of an example apparatus for calculating an internal resistance of a battery according to one embodiment of the present disclosure. Referring to FIG. 6, an apparatus for calculating an internal resistance of a battery according to the present embodiment may include a processor **600** and a memory **200**.

[0067] The processor **600** may be implemented as a central processing unit (CPU) or a system-on-chip (SoC), and may be configured to execute an operating system or application to control a plurality of hardware or software components connected to the processor **600**, and may be configured to process various pieces of data and perform calculations. The processor **600** may be configured to execute at least one instruction stored in the memory **200**, described below, and store execution result data in the memory **200**.

[0068] In the present embodiment, the processor **600** may be configured to calculate an internal resistance of a battery B (that is, a battery cell) mounted on a vehicle. Specifically, the processor **600** may collect a current parameter and a voltage parameter of the battery B when a data collection condition is satisfied, the data collection condition being predefined based on the discharge current characteristics of the battery while the vehicle is in a driving state, and the processor **600** may calculate the internal resistance of a battery cell based on the collected current parameter and voltage parameter. The processor **600** may be implemented as a battery management system (BMS) configured to collect a voltage, a current, a temperature, a state of charge (SOC), and state of a health (SOH) of a battery pack, a battery module, or a battery cell, and to perform battery charge/discharge control, monitoring and management, and cell balancing based on the collected data, or may be implemented as a microcontroller unit (MCU) in the BMS. The specific operation of the processor **600** will be divided into a first embodiment and a second embodiment in the following description.

[0069] The memory **200** may store at least one instruction to be executed by the processor **600**. The memory **200** may be implemented as a volatile storage medium and/or a non-volatile storage medium, for example, as a read-only memory (ROM) and/or a random access memory (RAM).

[0070] The operation of the processor **600** for calculating an internal resistance of a battery cell will be divided into the first embodiment and the second embodiment in the following detailed description based on the above content.

1. First Embodiment: Actual Driving State of Vehicle

[0071] FIG. 7 shows an example of a data collection condition in the apparatus for calculating an internal resistance of a battery according to one embodiment of the present disclosure. FIGS. 8 to 10 show a process for calculating an internal resistance in the apparatus for calculating an internal resistance of a battery according to one embodiment of the present disclosure.

[0072] As described above, the processor **600** may be configured to collect a current parameter and a voltage parameter of a battery when a data collection condition is satisfied, the data collection condition being predefined based on the discharge current characteristics of the battery while the vehicle is in a driving state, and the processor **600** may calculate an internal resistance of a battery cell based on the collected current parameter and voltage parameter.

[0073] In the first embodiment, the driving state may correspond to an actual driving state (for example, a state in which the vehicle is actually driving on a road) (whether the vehicle is in the actual driving state may be determined, for example, through a vehicle speed sensor or Global Positioning System (GPS) module mounted on the vehicle). That is, in the first embodiment, the processor **600** may be configured to calculate the internal resistance of the battery cell based on a current parameter and a voltage parameter collected while the vehicle is in the actual driving state while the battery is mounted on the vehicle. In addition, in the first embodiment, the current parameter of the battery may correspond to a discharge current of a battery pack, and the voltage parameter of the battery may correspond to a voltage of the battery cell. The collection of the discharge current of the battery pack and the voltage of the battery cell may be supported by a battery monitoring function of the processor **600** implemented as a BMS.

[0074] In addition, in the first embodiment, as shown in FIG. 7, a data collection condition serving as a start condition for calculating the internal resistance of the battery cell may be defined as a condition that a rate of change in discharge current of the battery pack is less than or equal to a preset threshold value during a preset reference time period while the vehicle is in the actual driving state. That is, in order to simulate a constant current discharge environment that is generally applied when an internal resistance of a single cell of the battery is measured, the processor **600** may be configured to capture a state in which an amount of change in discharge current of the battery pack is small while the vehicle is in the actual driving state. The reference time period and the threshold value adopted as references for determining “a state in which the change in the discharge current of the battery pack is small” may be preset in the processor **600** based on the implementation and experimental results (for example, reference value: 10 sec and threshold value: 0.3 A/sec). For a clear distinction between terms, the data collection condition applied in the first embodiment is referred to as a first data collection condition.

[0075] When the first data collection condition is satisfied, and thus a discharge current of the battery pack and a cell voltage of the battery cell are collected in the reference time period, the processor **600** may be configured to acquire an average value of the discharge current and an average value of the cell voltage in the reference time period and may calculate the internal resistance of the battery cell based on the acquired average value of the discharge current and the acquired average value of the cell voltage. For example, the processor **600** may be configured to calculate the internal resistance of the battery cell through a calculation method of dividing the average value of the cell voltage by the average value of the discharge current.

[0076] As described above, the internal resistance of the battery cell may be calculated through the calculation method of dividing the average value of the cell voltage by the average value of the discharge current. However, since an inherent error of an “average value” adopted as a factor in a process of calculating the internal resistance of the battery cell may cause an error in a calculated value of the internal resistance of the battery cell, in the present embodiment, the processor **600** may calculate the internal resistance of the battery pack through a method of acquiring an average value of a discharge current of the battery pack and an average value of a cell voltage of the battery cell in a plurality of time periods and interpolating the average values in the plurality of time periods, thereby compensating for an error of the internal resistance of the battery cell calculated in a single time period.

[0077] Specifically, first, the processor **600** may be configured to obtain a first average value of a discharge current of the battery and a first average value of a cell voltage of the battery in the first reference time period in which the first data collection condition is satisfied. The processor **600** may be configured to obtain a second average value of a discharge current of the battery and a second average value of a cell voltage of the battery in a second reference time period in which the first data collection condition is subsequently satisfied after the first reference time period.

[0078] Subsequently, the processor **600** may be configured to calculate the internal resistance of the battery cell through a method of interpolating the first average value of the discharge current, the first average value of the cell voltage, the second average value of the discharge current, and the second average value of the cell voltage. As an example of such an interpolation method, a linearization method, in which a straight line is used to connect a point (hereinafter referred to as a first point) defined by the first average value of the discharge current and the first average value of the cell voltage to a point (hereinafter referred to as a second point) defined by the second average values of the discharge current and the second average value of the cell voltage, may be applied. Accordingly, the processor **600** may be configured to calculate a slope (absolute value of the slope) of the straight line between the first point and the second point in a coordinate system, in which a horizontal axis and a vertical axis represent a current (battery pack current) and a voltage (battery cell voltage), respectively, as the internal resistance of the battery cell.

[0079] Above, an embodiment has been described in which the internal resistance of the battery

cell is calculated in at least two reference time periods required to compensate for an error of the internal resistance of the battery cell calculated in a single time period. However, accuracy in calculating a resistance of the battery cell may increase as the number of reference time periods for which an average value of a discharge current and an average value of a cell voltage are acquired increases, and thus the processor **600** may be configured to calculate an internal resistance of the battery cell in consideration of an average value of a discharge current and an average value of a cell voltage obtained for a total of four reference time periods, as shown in FIG. **8**, thereby improving accuracy in calculating the internal resistance.

[0080] Meanwhile, the processor **600** may be configured to calculate an internal resistance of the battery cell for each SOC of the battery pack and may derive a change in internal resistance according to the SOC of the battery pack. In parallel therewith, the processor **600** may be configured to calculate an internal resistance of the battery cell for each discharge rate (discharge C-rate) of the battery pack and may derive a change in internal resistance according to the discharge rate of the battery pack (such an operation is based on a state in which the SOC and the discharge rate of the battery pack have different values according to a time point at which the internal resistance of the battery cell is calculated). Collection of the SOC and the discharge rate of the battery pack may be supported by a battery monitoring function of the processor **600** implemented as a BMS.

[0081] In order to calculate the internal resistance of the battery cell for each SOC and each discharge rate of the battery pack, as shown as an example in Table 1 below, the processor **600** may be configured to first collect a battery cell voltage according to an SOC and a discharge rate of the battery pack. FIG. **9** is a graph showing collection results in Table 1.

TABLE-US-00001 TABLE 1 SOC 20% 50% 80% Discharge rate 0.11 C 3.59 V 3.72 V 3.94 V (C-rate) 0.33 C 3.57 V 3.69 V 3.93 V 0.66 C 3.55 V 3.66 V 3.91 V 1.00 C 3.52 V 3.64 V 3.89 V

[0082] Based on the data in Table 1, the processor **600** may be configured to calculate the internal resistance of the battery cell for each SOC of the battery pack. That is, the processor **600** may be configured to calculate the internal resistance of the battery cell for each SOC of the battery pack through a process of repeatedly performing a process of calculating a slope of a straight line according to the above-described interpolation method for each SOC. FIG. **10** shows an example in which, when the SOC of the battery pack are respectively 20%, 50%, and 80%, the internal resistances of the battery cell are calculated to be 0.0018, 0.0019, and 0.0014, respectively.

Thereafter, the processor **600** may be configured to derive a change in internal resistance according to the SOC of the battery pack (for example, relationship data between a change in SOC of the battery pack and a change in internal resistance of the battery cell) and may be configured to store the derived change in internal resistance in the memory **200**. The relationship data stored in the memory **200** may be used to predict and manage the lifespan of the battery pack later.

[0083] Likewise, the processor **600** may be configured to calculate the internal resistance of the battery cell for each discharge rate of the battery pack based on the data in Table 1. That is, the processor **600** may be configured to calculate the internal resistance of the battery cell for each discharge rate of the battery pack through a process of repeatedly performing a process of calculating a slope of a straight line according to the above-described interpolation method for each discharge rate. Thereafter, the processor **600** may be configured to derive a change in internal resistance according to the discharge rate of the battery pack (for example, relationship data between a change in discharge of the battery pack and a change in internal resistance of the battery cell) and may be configured to store the derived change in internal resistance in the memory **200**. The relationship data stored in the memory **200** may be used to predict and manage the lifespan of the battery pack later.

2. Second Embodiment: Simulated Driving State of Vehicle

[0084] The above-described first embodiment focuses on a configuration for starting a calculation operation of an internal resistance of the battery cell when the first data collection condition is

satisfied, namely that the rate of change in discharge current of the battery pack is less than or equal to the preset threshold value during the preset reference time period while the vehicle is in the actual driving state. However, in an actual driving environment of the vehicle, a case in which the above first data collection condition is satisfied may not occur for a long time (for example, a case in which the first data collection condition is satisfied may not occur for a preset time (for example, 72 hours)).

[0085] In order to calculate an internal resistance of the battery cell installed on the vehicle in such a situation, the second embodiment presents a configuration for calculating the internal resistance of the battery cell while the vehicle is in a simulated driving state rather than an actual driving state. That is, the second embodiment functions as a redundancy logic for the first embodiment. The simulated driving state may correspond, for example, to a state in which only a motor (driving motor) of the vehicle is driven without the actual driving of the vehicle in a state in which the vehicle is positioned on a chassis dynamometer, and it is based on assuming that the vehicle is positioned in a specific space (for example, a vehicle repair shop) in which the chassis dynamometer is installed.

[0086] Unlike the actual driving state, when the vehicle is in the simulated driving state, since the battery pack may be controlled to be discharged at a constant current, in the second embodiment, a data collection condition is not defined according to a rate of change in discharge current of the battery pack, but is defined as a condition that constant current discharging of the battery pack is performed. Accordingly, when constant current discharging of the battery pack is performed while the vehicle is in the simulated driving state, the processor **600** may be configured to determine that the data collection condition is satisfied. For a clear distinction between terms, the data collection condition applied in the second embodiment is referred to as a second data collection condition.

[0087] When the second data collection condition is satisfied, the processor **600** may be configured to collect a discharge constant current of the battery pack and a cell voltage of the battery as a current parameter and a voltage parameter of the battery and may be configured to calculate an internal resistance of the battery cell based on the collected discharge constant current and cell voltage. For example, the processor **600** may be configured to calculate the internal resistance of the battery cell through a calculation method of dividing the cell voltage by the discharge current.

[0088] As described above, the internal resistance of the battery cell may be calculated through the calculation method of dividing the cell voltage by the discharge current. However, as in the first embodiment, in order to improve accuracy in calculating the internal resistance of the battery cell, the processor **600** may be configured to acquire a discharge current of the battery pack and a cell voltage of the battery cell at a plurality of time points and may be configured to calculate an internal resistance of the battery cell through a method of interpolating a discharge current and a cell voltage for each of the plurality of time points, thereby compensating for an error of the internal resistance of the battery cell calculated for a single time point.

[0089] Specifically, the processor **600** may be configured to collect a first value of a discharge constant current of the battery and a first value of a cell voltage of the battery at a first reference time point at which the second data collection condition is satisfied. The processor **600** may be configured to collect a second value of a discharge constant current of the battery and a second value of a cell voltage of the battery at a second reference time at which the second data collection condition is subsequently satisfied after the first reference time.

[0090] Subsequently, the processor **600** may be configured to calculate the internal resistance of the battery cell through a method of interpolating the collected first value of the discharge constant current, the collected first value of the cell voltage, the collected second value of the discharge constant current, and the collected second value of the cell voltage. As an example of such an interpolation method, a linearization method, in which a straight line is used to connect a point defined by the first value of the discharge constant current and the first value of the cell voltage (hereinafter referred to as a first point which is distinguished from the first point defined in the first

embodiment) to a point defined by the second value of the discharge constant current and the second average value of the cell voltage (hereinafter referred to as a second point which is distinguished from the second point defined in the first embodiment), may be applied. Accordingly, the processor **600** may be configured to calculate a slope (absolute value of the slope) of the straight line between the first point and the second point in a coordinate system in which a horizontal axis and a vertical axis represent a current (battery pack current) and a voltage (battery cell voltage), respectively, as the internal resistance of the battery cell. Similar to the first embodiment, the processor **600** may be configured to calculate an internal resistance of the battery cell in consideration of discharge constant current values and cell voltage values obtained for three or more time points, thereby improving accuracy in calculating the internal resistance.

[0091] Meanwhile, the processor **600** may be configured to calculate the internal resistance of the battery cell for each SOC of the battery pack and derive a change in internal resistance according to the SOC of the battery pack. In parallel therewith, the processor **600** may be configured to calculate the internal resistance of the battery cell according to a discharge rate (discharge C-rate) of the battery pack and derive a change in internal resistance according to the discharge rate of the battery pack (such an operation is based on a state in which the SOC and the discharge rate of the battery pack have different values according to a time point at which the internal resistance of the battery cell is calculated). Collection of the SOC and the discharge rate of the battery pack may be supported by the battery monitoring function of the processor **600** implemented as a BMS. Such a configuration is the same as the first embodiment, and thus a detailed description thereof will be omitted.

3. Method of Calculating Internal Resistance of Battery

[0092] FIGS. **11** to **13** are a flowchart of an example method of calculating an internal resistance of a battery according to one embodiment of the present disclosure. The method of calculating an internal resistance of a battery performed by the processor **600** will be described with reference to FIGS. **11** to **13**. Detailed description of content that overlaps the above-described content will be omitted, and the description will focus on a time-series configuration.

[0093] First, referring to FIG. **11**, the method of calculating an internal resistance of a battery of the present embodiment may include operation **S100** of determining whether a data collection condition is satisfied, the data collection condition being predefined based on the discharge current characteristics of a battery while a vehicle is in a driving state, operation **S200** of, when the data collection condition is satisfied, collecting a current parameter and a voltage parameter of the battery, and operation **S300** of calculating an internal resistance of the battery based on the current parameter and voltage parameter collected in operation **S200**. Operation **S100** includes operations **S110** and **S120** described below, operation **S200** includes operations **S210** to **S230**, described below, and operation **S300** includes operations **S310** to **S330**.

[0094] In operations **S100** to **S300**, the processor **600** operates differently depending on whether “a condition that a rate of change in discharge current of the battery is less than or equal to a preset threshold value during a preset reference time period while the vehicle is in an actual driving state” is satisfied. That is, when the first data collection condition is satisfied, the internal resistance of the battery cell is calculated according to the first embodiment described above. When the first data collection condition is not satisfied, the internal resistance of the battery cell is calculated according to the second embodiment, which serves as a redundancy logic for the first embodiment. A time-series configuration in which the first or second embodiment is performed according to whether the first data collection condition is satisfied will be described below.

[0095] Referring to FIG. **12**, the processor **600** preferentially determines whether the rate of change in discharge current of the battery is less than or equal to the preset threshold value during the preset reference time period while the vehicle is in the actual driving state (**S110**). That is, the processor **600** preferentially determines whether the first data collection condition is satisfied.

[0096] When it is determined in operation **S110** that the first data collection condition is satisfied,

the processor **600** collects a discharge current and a cell voltage of the battery as a current parameter and a voltage parameter of the battery in the first and second time periods in which the first data collection condition is satisfied (**S210**).

[0097] Subsequently, the processor **600** acquires a first average value of the discharge current of the battery and a first average value of the cell voltage of the battery in the first reference time period and acquires a second average value of the discharge current of the battery and a second average value of the cell voltage of the battery cell in the second reference time period (**S310**). Then, the processor **600** calculates the internal resistance of the battery through a method of interpolating the first average value of the discharge current, the first average value of the cell voltage, the second average value of the discharge current, and the second average value of the cell voltage (**S320**). In this case, the processor **600** may calculate a slope of a straight line, which connects a point defined by the first average value of the discharge current and the first average value of the cell voltage and a point defined by the second average value of the discharge current and the second average value of the cell voltage, as the internal resistance of the battery.

[0098] Meanwhile, when the internal resistance of the battery is calculated for each SOC of the battery as operations **S110**, **S210**, **S310**, and **S320** are repeatedly performed, the processor **600** may operate to derive a change in internal resistance according to the SOC of the battery and store the derived change in internal resistance in a memory **200**. Meanwhile, when the internal resistance of the battery is calculated for each discharge rate of the battery as operations **S110**, **S210**, **S310**, and **S320** are repeatedly performed, the processor **600** may operate to derive a change in internal resistance according to the discharge rate of the battery and store the derived change in internal resistance in the memory **200**.

[0099] Next, referring to FIG. 13, when it is determined in operation **S110** that the first data collection condition is not satisfied, the processor **600** determines whether constant current discharging of the battery is performed while the vehicle is in a simulated driving state (**S120**). That is, when the first data collection condition is not satisfied, the processor **600** determines whether a second data collection condition is satisfied, the second data collection condition serving as a condition for calculating the internal resistance of the battery according to a redundancy logic of the first data collection condition being not satisfied.

[0100] When it is determined in operation **S120** that the second data collection condition is satisfied, the processor **600** collects a discharge current and a cell voltage of the battery as a current parameter and a voltage parameter of the battery at the first and second reference time points at which the second data collection condition is satisfied (**S220** and **S230**). That is, the processor **600** collects a first value of a discharge constant current and a first value of a cell voltage of the battery at the first reference time point at which the second data collection condition is satisfied (**S220**) and collects a second value of a discharge constant current and a second value of a cell voltage of the battery at the second reference time point at which the second data collection condition is satisfied after the first reference time point (**S230**).

[0101] Subsequently, the processor **600** may calculate the internal resistance of the battery cell through a method of interpolating the first value of the discharge constant current, the first value of the cell voltage, the second value of the discharge constant current, and the second value of the cell voltage. In this case, the processor **600** may calculate a slope of a straight line, which connects a point defined by the first value of the discharge constant current and the first value of the cell voltage and a point defined by the second value of the discharge constant current and the second value of the cell voltage, as the internal resistance of the battery.

[0102] Meanwhile, when the internal resistance of the battery is calculated for each SOC of the battery as operations **S120**, **S220**, **S230**, and **S330** are repeatedly performed, the processor **600** may operate to derive a change in internal resistance according to the SOC of the battery and store the derived change in internal resistance in the memory **200**. In order to differentially form an SOC of the battery, the processor **600** may form a plurality of SOC's through a method of calculating a

battery charging target current for satisfying a plurality of SOC's input by an operator, and performing feedback control for allowing a currently input current of a battery pack to follow the battery charging target current. Similarly, when the internal resistance of the battery is calculated for each discharge rate of the battery as operations **S120**, **S220**, **S230**, and **S330** are repeatedly performed, the processor **600** may operate to derive a change in internal resistance according to the discharge rate of the battery and store the derived change in internal resistance in a memory **200**.

[0103] The method of calculating an internal resistance of a battery according to the present embodiment can be implemented as a non-transitory computer-readable medium having instructions stored thereon for executing operations **S100** to **S300** described above suitable for use with a general-purpose digital computer that executes the instructions. The non-transitory computer-readable medium may correspond to a ROM, a RAM, a magnetic medium such as a hard disk, a floppy disk, or magnetic tape, an optical medium such as a compact disc (CD)-ROM or a digital video disc (DVD), a magneto-optical medium such as a floptical disk, or a hardware device specially configured to store and execute program instructions such as a flash memory.

[0104] In this way, according to the present disclosure, when a predefined data collection condition is satisfied based on the discharge current characteristics of a battery while a vehicle is in a driving state, by adopting a method of calculating an internal resistance of a battery based on a current parameter and a voltage parameter of a battery, even in a state in which a battery is mounted on the vehicle, the internal resistance of a battery cell can be measured without the separation or removal of the battery from the vehicle.

[0105] However, effects that can be achieved through the present disclosure are not limited to the above-described effects and other effects that are not described may be clearly understood by those skilled in the art from the detailed descriptions.

[0106] Herein, the embodiments described herein may be implemented, for example, as a method or process, a device, a non-transitory computer readable medium (e.g., having instructions described herein stored thereon), a data stream, or a signal. Although discussed in the context of a single type of implementation (for example, discussed only as a method), features discussed herein may also be implemented in other forms (for example, a device or a program). The device may be implemented by suitable hardware, software, firmware, and the like. The method may be implemented on a device, such as a processor that generally refers to a processing device including a computer, a microprocessor, an integrated circuit, a programmable logic device, etc. The processor includes a communication device such as a computer, a cell phone, a personal digital assistant (PDA), and other devices that facilitate communication of information between the device and end-users.

[0107] Although the present disclosure has been described with reference to embodiments and drawings illustrating aspects thereof, the present disclosure is not limited thereto. Various modifications and variations can be made by a person skilled in the art to which the present disclosure belongs within the scope of the technical spirit of the present disclosure and the claims and their equivalents, below.

Claims

1. An apparatus for calculating an internal resistance of a battery, the apparatus comprising: a processor; and a memory configured to store instructions for execution by the processor, wherein, when executing the instructions, the processor is configured to calculate an internal resistance of a battery mounted on a vehicle by: when the processor determines that a data collection condition is satisfied, collecting a current parameter and a voltage parameter of the battery and calculating the internal resistance of the battery based on the current parameter and the voltage parameter, wherein the data collection condition is predefined based on discharge current characteristics of the battery while the vehicle is in a driving state.

2. The apparatus of claim 1, wherein the data collection condition is that a rate of change in discharge current of the battery is less than or equal to a preset threshold value during a preset reference time period while the vehicle is in an actual driving state.
3. The apparatus of claim 2, wherein, when executing the instructions, the processor is configured to collect a discharge current of the battery as the current parameter and a cell voltage of the battery as the voltage parameter during the preset reference time period in which the data collection condition is satisfied, and calculate the internal resistance of the battery based on the discharge current and the cell voltage.
4. The apparatus of claim 3, wherein, when executing the instructions, the processor is configured to acquire an average value of the discharge current and an average value of the cell voltage in the preset reference time period and calculate the internal resistance of the battery based on the average value of the discharge current and the average value of the cell voltage.
5. The apparatus of claim 4, wherein, when executing the instructions, the processor is configured to acquire a first average value of the discharge current of the battery and a first average value of the cell voltage of the battery in a first reference time period in which the data collection condition is satisfied, acquire a second average value of the discharge current of the battery and a second average value of the cell voltage of the battery in a second reference time period in which the data collection condition is satisfied and which is after the first reference time period, and calculate the internal resistance of the battery by interpolating the first average value of the discharge current, the first average value of the cell voltage, the second average value of the discharge current, and the second average value of the cell voltage.
6. The apparatus of claim 5, wherein, when executing the instructions, the processor is configured to calculate a slope of a straight line as the internal resistance of the battery, the straight line connecting a first point defined by the first average value of the discharge current and the first average value of the cell voltage to a second point defined by the second average value of the discharge current and the second average value of the cell voltage.
7. The apparatus of claim 1, wherein the data collection condition is that constant current discharging of the battery is performed while the vehicle is in a simulated driving state.
8. The apparatus of claim 7, wherein, when executing the instructions, the processor is configured to, when the data collection condition is satisfied, collect a discharge constant current of the battery as the current parameter and a cell voltage of the battery as the voltage parameter and calculate the internal resistance of the battery based on the discharge constant current and the cell voltage.
9. The apparatus of claim 8, wherein, when executing the instructions, the processor is configured to collect a first value of the discharge constant current and a first value of the cell voltage of the battery at a first reference time point at which the data collection condition is satisfied, collect a second value of the discharge constant current and a second value of the cell voltage of the battery at a second reference time point at which the data collection condition is satisfied and which is after the first reference time point, and calculate the internal resistance of the battery by interpolating the first value of the discharge constant current, the first value of the cell voltage, the second value of the discharge constant current, and the second value of the cell voltage.
10. The apparatus of claim 9, wherein, when executing the instructions, the processor is configured to calculate a slope of a straight line as the internal resistance of the battery, the straight line connecting a first point defined by the first value of the discharge constant current and the first value of the cell voltage to a second point defined by the second value of the discharge constant current and the second value of the cell voltage.
11. The apparatus of claim 1, wherein, when executing the instructions, the processor is configured to calculate the internal resistance of the battery for each state of charge (SOC) of the battery and derive a change in internal resistance according to the SOC of the battery.
12. The apparatus of claim 1, wherein, when executing the instructions, the processor is configured to calculate the internal resistance of the battery for each discharge rate of the battery and derive a

change in internal resistance according to the discharge rate of the battery.

13. A method of calculating an internal resistance of a battery mounted on a vehicle, the method comprising: determining, by a processor, whether a data collection condition is satisfied, wherein the data collection condition is predefined based on discharge current characteristics of the battery when the vehicle is in a driving state; when the processor determines that the data collection condition is satisfied, collecting, by the processor, a current parameter and a voltage parameter of the battery; and calculating, by the processor, the internal resistance of the battery based on the current parameter and the voltage parameter.

14. The method of claim 13, wherein the determining comprises, when a rate of change in discharge current of the battery is less than or equal to a preset threshold value during a preset reference time period while the vehicle is in an actual driving state, the processor determining that the data collection condition is satisfied.

15. The method of claim 14, wherein the collecting comprises, in a first reference time period in which the data collection condition is satisfied and in a second reference time period in which the data collection condition is satisfied and which is after the first reference time period, the processor collecting a discharge current of the battery as the current parameter and a cell voltage of the battery as the voltage parameter, and the calculating comprises the processor acquiring an average value of the discharge current and an average value of the cell voltage for the first and second reference time periods and then calculating the internal resistance of the battery based on the average value of the discharge current and the average value of the cell voltage.

16. The method of claim 15, wherein the calculating comprises the processor acquiring a first average value of the discharge current of the battery and a first average value of the cell voltage of the battery in the first reference time period, acquiring a second average value of the discharge current of the battery and a second average value of the cell voltage of the battery in the second reference time period, and calculating the internal resistance of the battery by interpolating the first average value of the discharge current, the first average value of the cell voltage, the second average value of the discharge current, and the second average value of the cell voltage.

17. The method of claim 13, wherein the determining comprises, when constant current discharging of the battery is performed while the vehicle is in a simulated driving state, the processor determining that the data collection condition is satisfied.

18. The method of claim 17, wherein the collecting comprises, when the data collection condition is satisfied, the processor collecting a discharge constant current of the battery as the current parameter and a cell voltage of the battery as the voltage parameter, and the calculating comprises the processor calculating the internal resistance of the battery based on the discharge constant current and the cell voltage.

19. The method of claim 18, wherein the collecting comprises the processor collecting a first value of the discharge constant current and a first value of the cell voltage of the battery at a first reference time point at which the data collection condition is satisfied, and collecting a second value of the discharge constant current and a second value of the cell voltage of the battery at a second reference time point at which the data collection condition is satisfied and which is after the first reference time point, and the calculating comprises the processor calculating the internal resistance of the battery by interpolating the first value of the discharge constant current, the first value of the cell voltage, the second value of the discharge constant current, and the second value of the cell voltage.

20. A non-transitory computer-readable medium having instructions stored thereon that, when executed by a processor, cause the processor to calculate an internal resistance of a battery mounted on a vehicle by performing a method comprising: determining whether a data collection condition is satisfied, wherein the data collection condition is predefined based on discharge current characteristics of the battery while the vehicle is in a driving state; when the data collection condition is satisfied, collecting a current parameter and a voltage parameter of the battery; and

calculating the internal resistance of the battery based on the current parameter and voltage parameter.
